Jane Illarionova

206-877-2871 | illariojane@gmail.com | JaneIllario | Jane Illarionova

EXPERIENCE

Microsoft – Software Engineer II - Microsoft Cloud Licensing Aug. 2023 – Present
Microsoft – Software Engineer - Microsoft Cloud Licensing Aug. 2021 – Aug. 2023

- Migrated 24B+ consumer entitlements (~40TB across 3.8B users) from an unstable SQL platform to Azure CosmosDB, architecting and shipping the large documents serialization solution, and driving the cost-improvements that slashed daily run cost by 86% while sustaining 99.9% QoS — eliminating recurring outages that impacted 110,000 customers and ~$700K in revenue.
- Led cutover for M365 Family Sharing, mirroring ~21M Consumer Group relationships to the new storage solution without service interuption. Removed dependency call-path, and ran cross-team standups and partner comms.
- Designed and operationalized AI agents and reusable skill frameworks for migration reconciliation, standardizing investigation workflows and amplifying team velocity through shared, automated debugging patterns.
- Architected and authored the Microsoft Cloud Licensing Reconciliation and AutoFix Worker spanning 3 cross-company datastores, built to scale with the growth of the with the commercial licensing system, defined the self-healing behaviour for all discrepancy types, and drove production, multi-cloud readiness.

Microsoft – Software Engineering Intern - Consumer Entitlements and Storage June 2020 – Sept. 2020

- Built a .NET Standard library for Azure App Configuration enabling dynamic, deployment-independent config updates across legacy and AKS. Drove cross-team adoption, establishing a shared standard for live configuration updates.

RBC Capital Markets – AI Engineering Intern - Aiden Lab Sept. 2019 – June 2020

- Engineered Python data pipelines for live and historical market signals across 200+ inputs, and tuned the deep reinforcement learning algorithm powering RBC's electronic trading platform, Aiden, to minimize VWAP slippage.

IMC Trading – Software Engineering Intern - Trade Execution June 2019 – Sept. 2019

- Optimized trade execution by developing high-performance C++ for FPGA Control and TCP Networking

Structural Genomics Consortium – AI Researcher - Epigenetics Drug Discovery Oct. 2018 – Mar. 2019

- Built and benchmarked PyTorch-based machine learning models for protein–ligand binding prediction in epigenetics drug discovery, transforming large-scale structural and drug interaction datasets into models that accelerate candidate evaluation versus decade-long manual processes.

EDUCATION

University of Toronto – Bachelor of Applied Science & Engineering 2021

- Specialization in Computer Engineering, Minor in Artificial Intelligence
- **Awards**: Certificate in Engineering Leadership, Dean's Honour List, Dean's Merit List, Engineering Society Award
- **Coursework**: Data Structures, Algorithms, Operating Systems, Machine Learning, Computer Security

PROJECTS

AuToronto

- Led health-monitoring systems engineering team activating and testing hardware components; contributed to ML model development and tuning for object-tracking. 1st Place two years consecutively in international competition.

BR(AI)N

- Designed and trained deep Convolutional Neural Networks for brain cancer diagnosis (performing with 96% accuracy) and implemented CUDA-parallelized Python to accelerate GPU-based image recognition.

SKILLS

Languages and Tools: C/C++, C#, .NET, Python, SQL, Kubernetes, Git, Linux, Docker, Azure, Kusto

Skills: Reinforcement Learning, Deep Learning, Data Engineering, Agentic Development, Cloud Architecture, Neural Networks, Distributed Systems

Duolingo Streak: 1318 days